

LSTM Networks with CTC Loss Applied to Gesture Typing Decoding

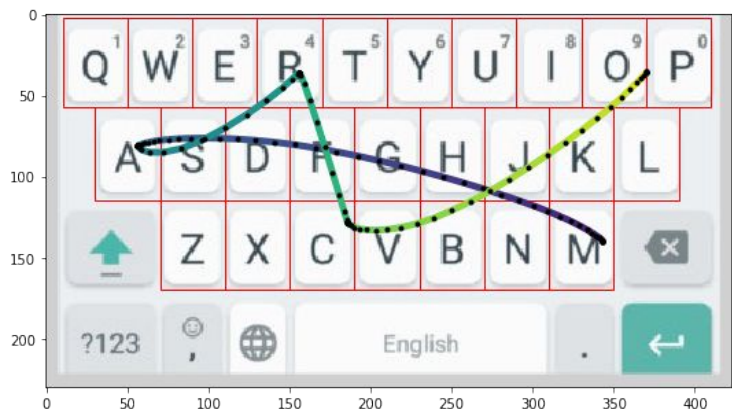
Willians Cassiano de Freitas Abreu

Outline

- Introduction
- Theoretical Background
- Gesture synthesis and decoding
- Experiments and results
- Conclusion

Introduction

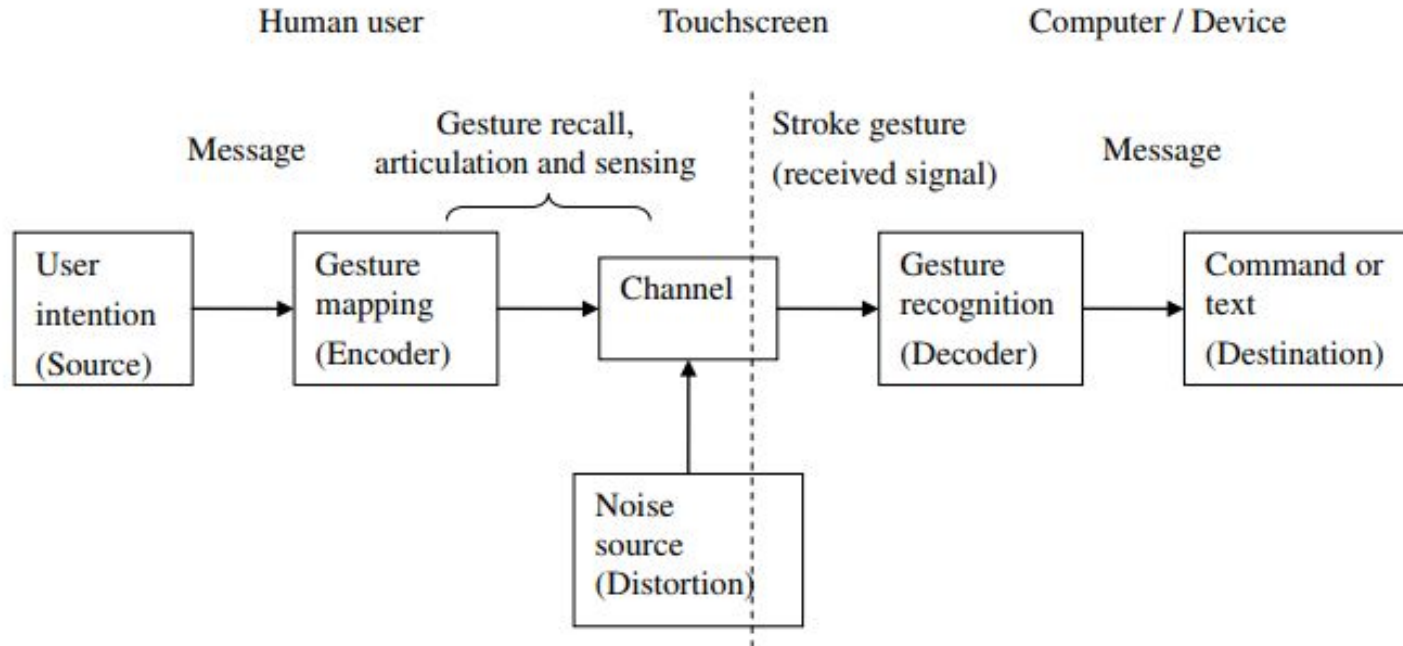
Problem Definition



MARCO

Introduction

Problem Definition

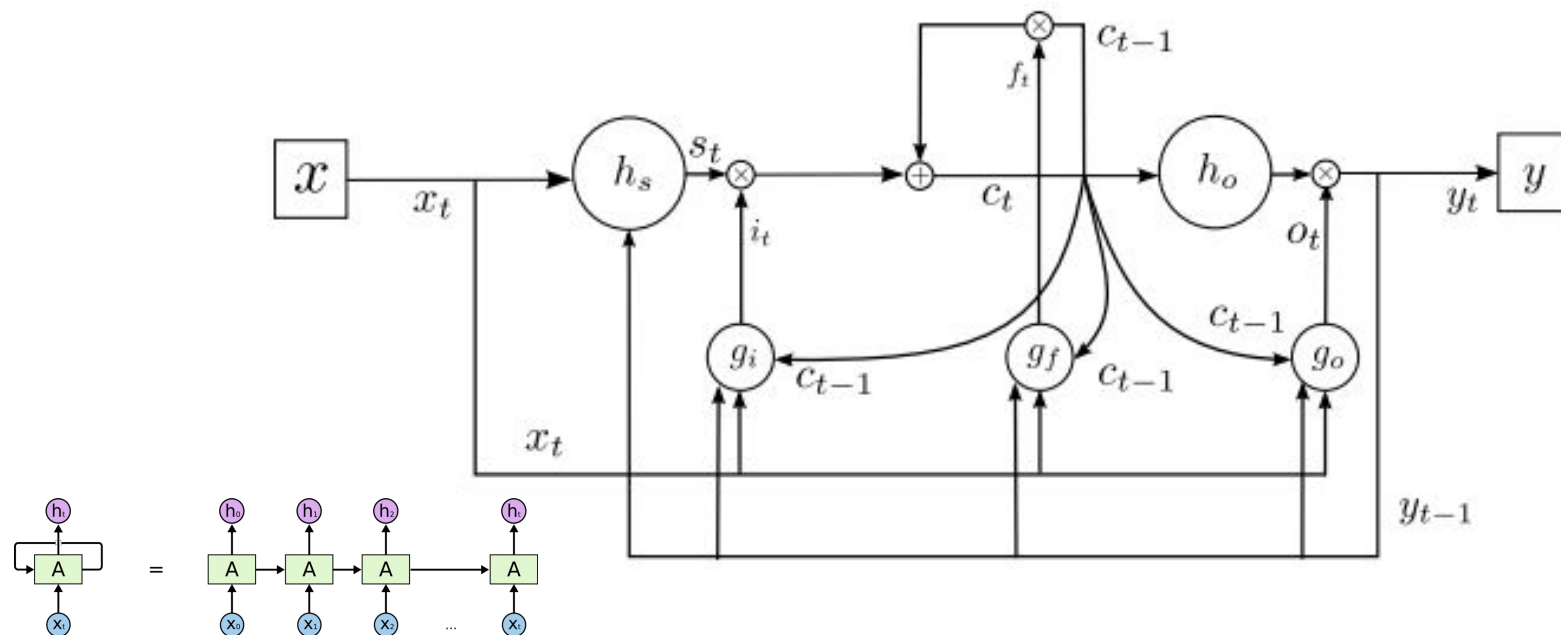


Motivation

Theoretical Background

Theoretical Background

LSTM

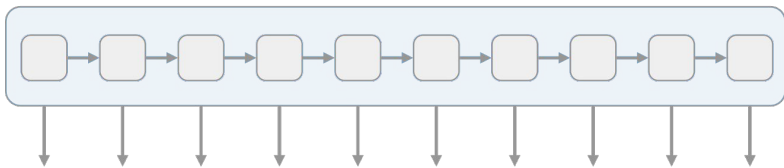


Theoretical Background

CTC



We start with an input sequence, like a spectrogram of audio.



The input is fed into an RNN, for example.

h	h	h	h	h	h	h	h	h	h
e	e	e	e	e	e	e	e	e	e
l	l	l	l	l	l	l	l	l	l
o	o	o	o	o	o	o	o	o	o
€	€	€	€	€	€	€	€	€	€

The network gives $p_t(a | X)$, a distribution over the outputs $\{h, e, l, o, \epsilon\}$ for each input step.

Theoretical Background

CTC

h	e	€			€			o	o
h	h	e			€	€		€	o
€	e	€			€	€		o	o

With the per time-step output distribution, we compute the probability of different sequences

h	e			o
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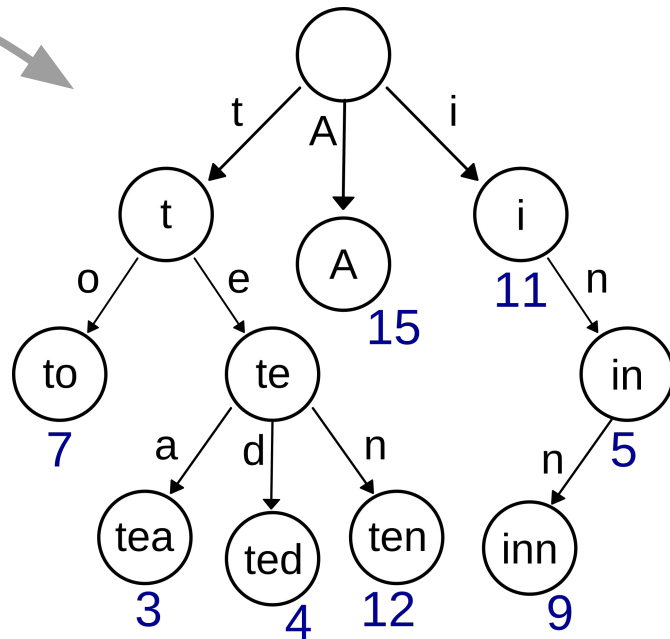
By marginalizing over alignments, we get a distribution over outputs.

$$P(w \mid G) = \sum_{A \in A_w} \prod_{t=1}^T p_t(c \mid G)$$

Theoretical Background

Trie - FST

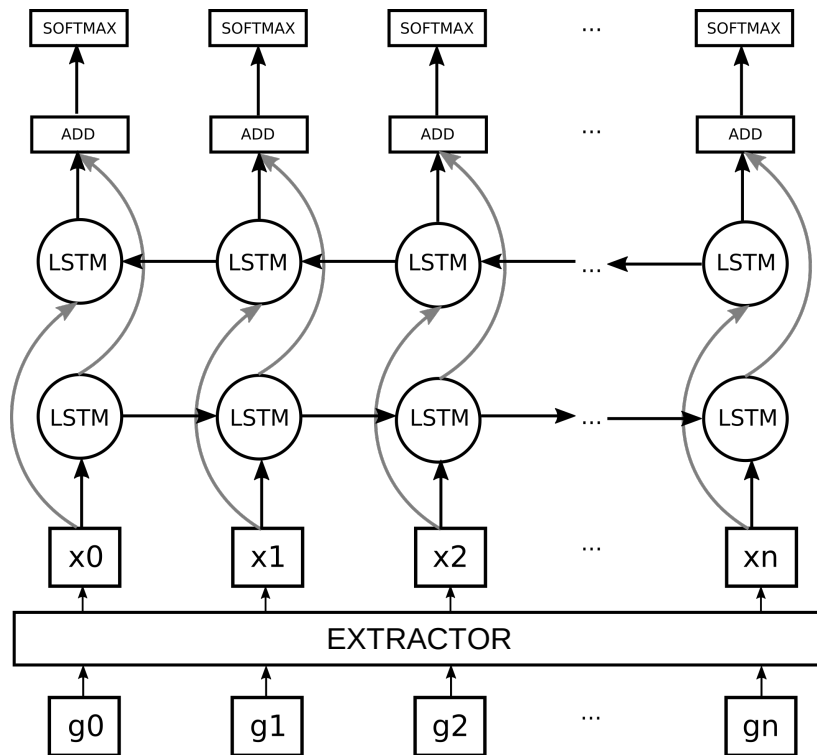
h	h	h	h	h	h	h	h	h	h
e	e	e	e	e	e	e	e	e	e
l	l	l	l	l	l	l	l	l	l
o	o	o	o	o	o	o	o	o	o
€	€	€	€	€	€	€	€	€	€



Gesture Synthesis and Decoding

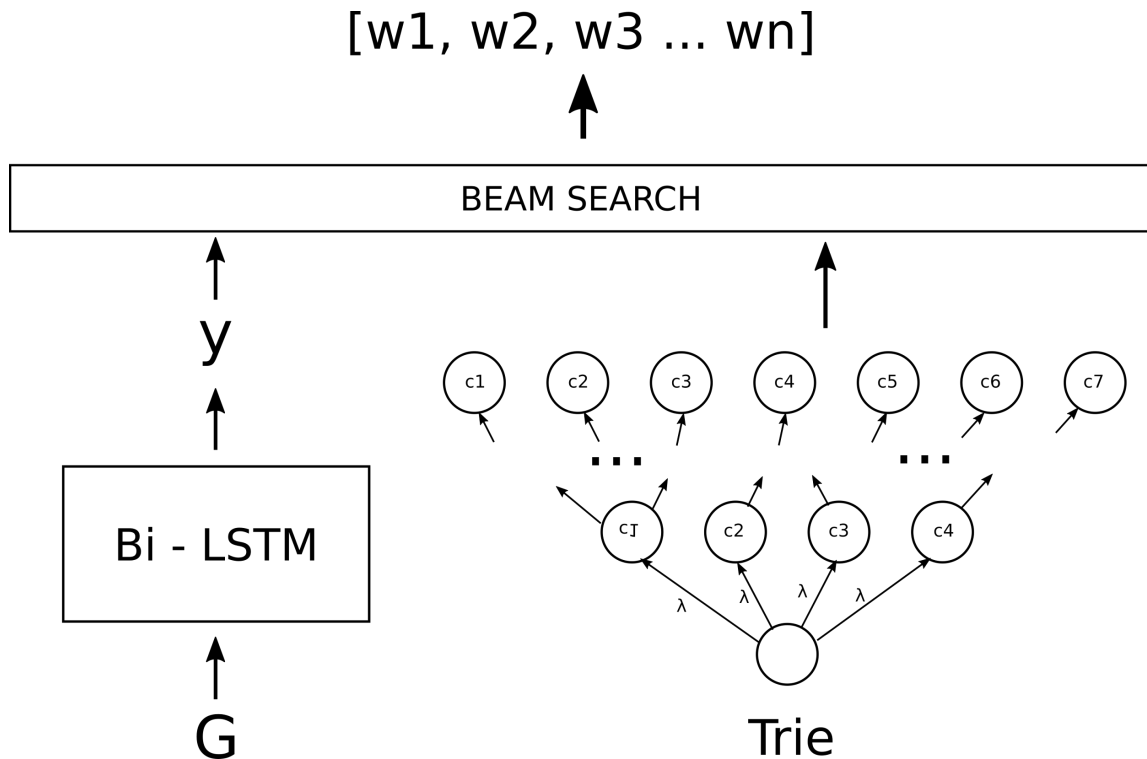
Decoding

Bi-LSTM



Decoding

Bi-LSTM + Trie



Synthesis

Algorithm

$w \leftarrow \text{word}$

$K \leftarrow m(\omega_{c_i})$ for c_i in w

$O \leftarrow \mathcal{N}(K, s_1)$

$\theta \leftarrow \mathcal{N}(0, s_2)$

$V \leftarrow ((\mathbf{o}_{i+1} - \mathbf{o}_i)/2)R(\theta) + \mathbf{o}_i$ for i from 1 to $|O| - 1$

$Z \leftarrow \text{interleave}(O, V)$

$G \leftarrow \text{mjt}(Z_i, Z_{i+1}, Z_{i+2})$ for i from 1 to $|Z| - 2$, with step 2

Synthesis

Algorithm

$w \leftarrow \text{word}$

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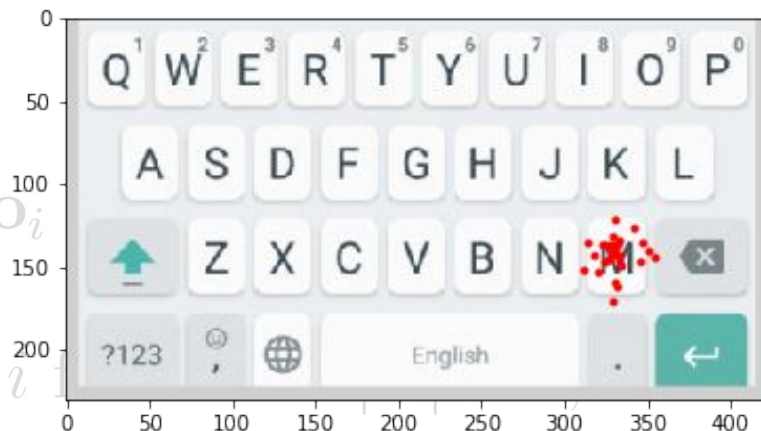
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step 2

Synthesis

Algorithm

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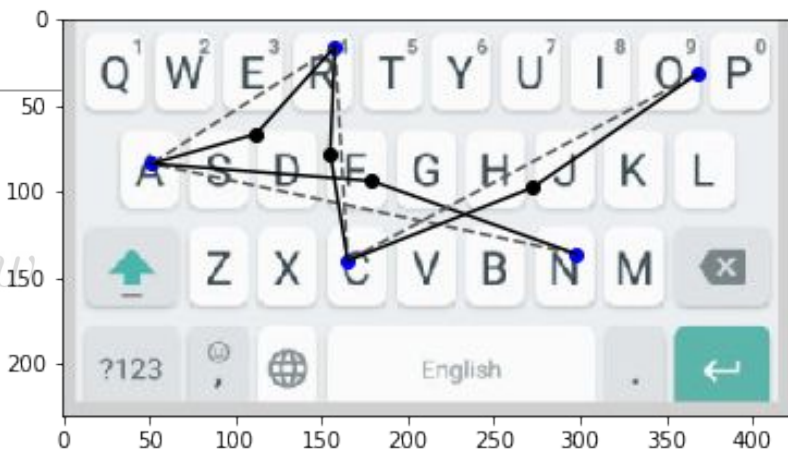
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Synthesis

Minimum Jerk Trajectory

$$C = \frac{1}{2} \int_{t_0}^{t_f} \left[\left(\frac{d^3 x}{dt^3} \right)^2 + \left(\frac{d^3 y}{dt^3} \right)^2 \right] dt$$

Synthesis

Algorithm

$w \leftarrow \text{word}$

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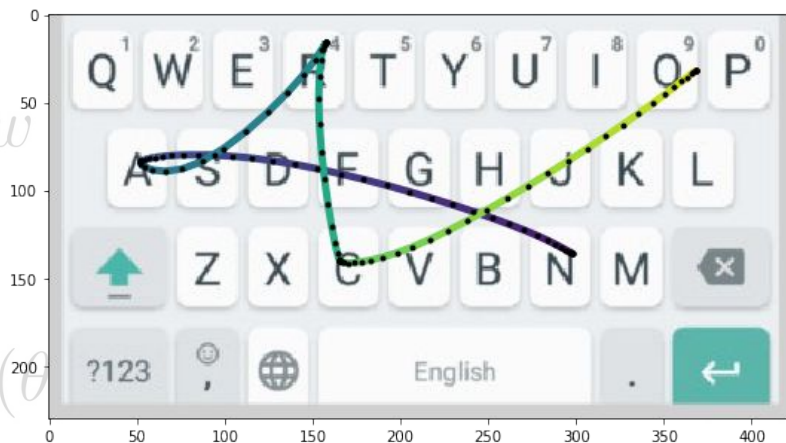
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Decoding

Feature Extraction



Original Proposal (Alsharif et al.)

Decoding

Feature Extraction



Our Proposal

$$X_{i,t} = P(\omega_i \mid g_t)$$

$$P(\omega_i \mid g_t) = \frac{1}{e^{\|m(\omega_i) - g_t\| \delta}}$$

Experiments and Results

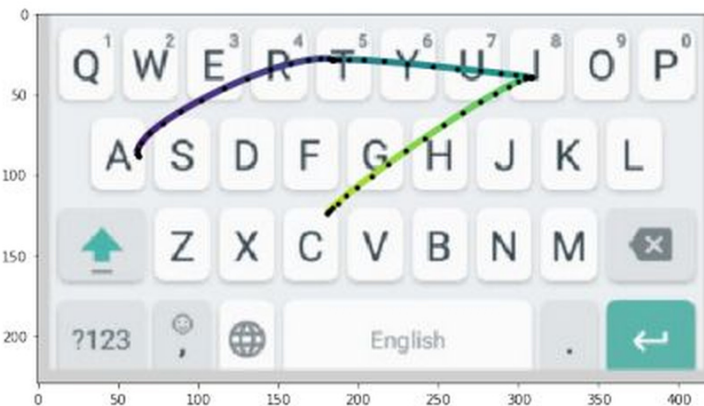
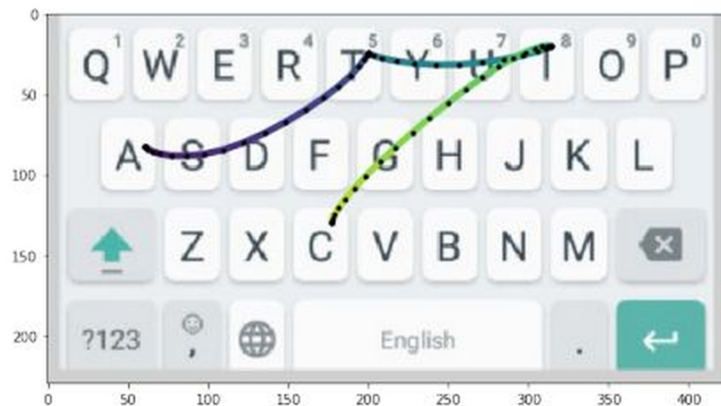
Results

Evaluation

Decoding method	Accuracy
argmax	65.51%
Trie	96.95%
Trie-3	97.76%

Experiments

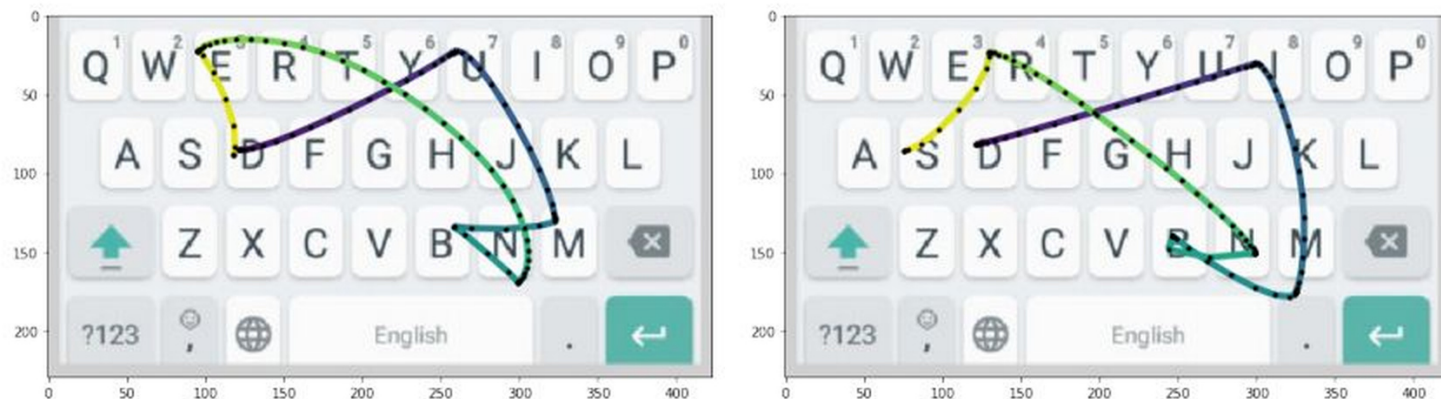
Word Recognition



- Example trajectories for the word **ATTIC**. In the example on the left, the model decodes it correctly. In the one on the right, the recognized word was **SATIN**.

Experiments

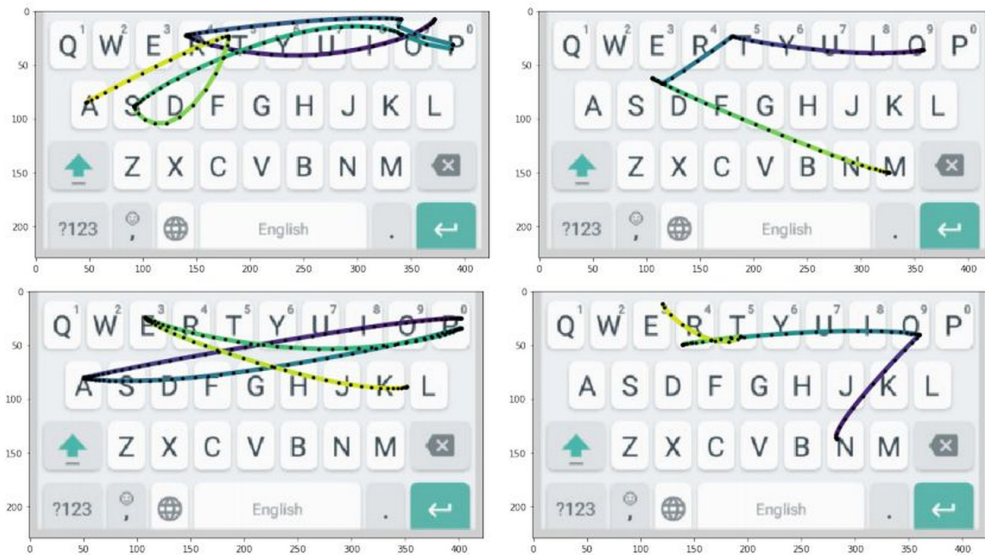
Word Recognition



- Example trajectories for the word DUMBNESS. In the example on the left, the model decodes it correctly. In the one on the right, the recognized word was DIMNESS.

Experiments

Transfer Learning to Portuguese



- Example trajectories for Portuguese words that were correctly classified. Clockwise, starting at the top left: PROPOSTA, ORDEM, PAPEL and NORTE.

Experiments

Comparison

Table 2 – Comparison with the *baseline*

	ALSHARIF et al.	Our model
LSTM dimension	400	512
Number of parameters	1.5M	2.2M
Feature extraction	No	Yes
Unique words	8k	62k
Training set	124k	150k
Test set	14k	12k
Accuracy (%)	93.5	96.9

Conclusion and Future Work

Thank You